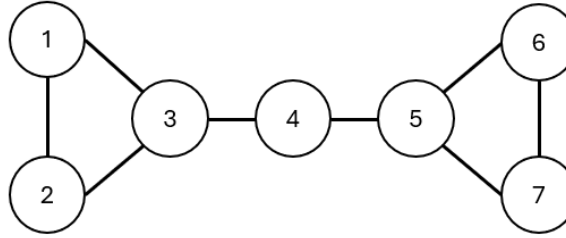


MID-TERM EXAM SOLUTION
(GRAPH MINING – SPRING 2026)

Question 1: Degree Centrality and Degree Distribution (10 points)

Consider the undirected graph G with nodes



Given:

- **Unnormalized Degree Centrality:** $C_D(v) = \sum_{j=1}^{|V|} A_{vj}$

Where:

- $C_D(v)$ is the degree centrality of node v .
- $\deg(v)$ is the total number of edges incident to node v .
- $|V|$ (often denoted as N) is the total number of nodes (vertices) in the graph.
- A is the adjacency matrix of the graph, where $A_{vj} = 1$ if there is an edge between node v and node j , and 0 otherwise.

- **Degree distribution** $p_k: p_k = \frac{N_k}{|V|}$

Where

- N_k is the total count of nodes with degree k
- $|V|$ is the total number of nodes in the network.

- a) Compute the **Degree centrality** of every node and then compute the **Degree distribution** of the graph. (7 points)
- b) One student claims that: “A node with the highest degree centrality is always the most important node in a network”. What is your opinion about this statement? (3 points)

Solution:

a)

- Degree centrality

Node	Neighbors	Degree ($C_D(v)$)
1	2, 3	2
2	1, 3	2
3	1, 2, 4	3
4	3, 5	2
5	4, 6, 7	3
6	5, 7	2
7	5, 6	2

- Degree Distribution

Degree 2: Nodes 1, 2, 4, 6, 7 $\rightarrow N_2 = 5$

Degree 3: Nodes 3, 5 $\rightarrow N_3 = 2$

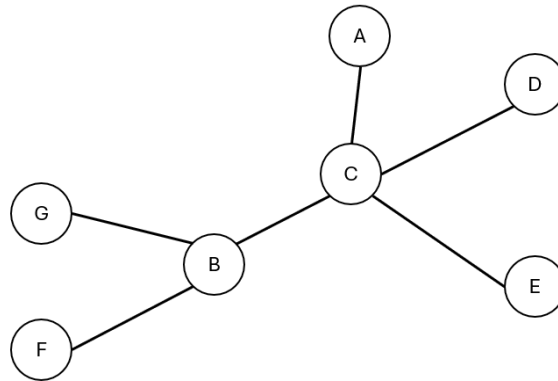
$$p_2 = \frac{5}{7}, p_3 = \frac{2}{7}$$

All other $p_k = 0$

- b) The statement is **incorrect**. Degree centrality measures only the number of direct connections a node has. However, a node's importance in a network depends on broader structural roles rather than just local connections. A node with fewer connections can be more critical if it lies on many shortest paths, which indicates high betweenness centrality, or connects different communities. Other centrality measures, such as eigenvector and PageRank, capture influence and importance more effectively in many real-world networks.

Question 2: Betweenness Centrality and Closeness Centrality (10 points)

Given an undirected graph G of 7 nodes.



Known that:

- **Betweenness centrality $C_B(v)$:**

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

Where:

- s and t are pairs of nodes in the network.
 - σ_{st} is the total number of shortest paths from the node s to node t .
 - $\sigma_{st}(v)$ is the number of those shortest paths that explicitly pass through the v .
 - If $s = t$, $\sigma_{st} = 1$, and if $v \in \{s, t\}$, $\sigma_{st}(v) = 0$.
- **Closeness Centrality (Normalized)**

$$C'_C(v) = \frac{|V| - 1}{\sum_{u \neq v} d(v, u)}$$

Where:

- $d(v, u)$ is the shortest path **distance** between node v and node u .

- Calculate the **Betweenness Centrality** and **Closeness Centrality** of Node C (7 points)
- If a company wants to identify the employee who:
 - best positioned to connect different departments, or
 - can quickly spread information for the whole organization

Which centrality measure is most appropriate for each goal? Explain your answer.

(3 points)

Solution:

a)

Observation: This is a tree, so there is exactly one shortest path between every pair.

→ Remove node C, then count components

Component	Nodes	Size
1	A	1
2	D	1
3	E	1
4	B, F, G	3

Count pairs that must pass through C:

- Between small components: A–D, A–E, D–E → 3 pairs
- Between small components and big components:
 - A ↔ (B, F, G): 3 pairs
 - D ↔ (B, F, G): 3 pairs
 - E ↔ (B, F, G): 3 pairs

Total: 3 + 9 = 12

Betweenness Centrality (Unnormalized): $C_B(C) = 12$

Normalized Betweenness:

$$\text{Normalization factor} = \frac{(n-1)(n-2)}{2} = \frac{6 \cdot 5}{2} = 15$$

$$C_B^{\text{norm}}(C) = \frac{12}{15} = 0.8$$

Closeness Centrality of Node C

- Compute Distances from C:

Node	Distance from C
A	1
B	1
D	1
E	1
F	2
G	2

Sum of distances: 1+1+1+1+2+2=8

Closeness Centrality

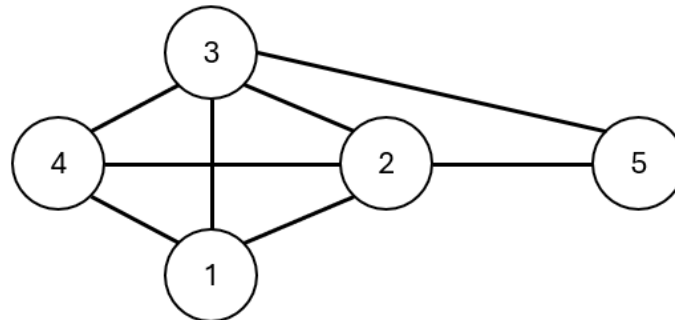
$$C_c(C) = \frac{6}{8} = 0.75$$

- b) To find the employee best suited to **connect different departments**, **betweenness centrality** is the best measure. It shows how often a node appears on the shortest paths between other nodes. An employee with high betweenness often serves as a bridge between groups or departments. If you remove or isolate such a person, it would greatly disrupt communication across the organization.

To identify the employee who can **quickly spread information throughout the organization**, **closeness centrality** is the best measure. It measures how close a node is to all other nodes based on shortest-path distances. An employee with high closeness can reach others quickly and effectively. This makes them ideal for spreading information rapidly throughout the network.

Question 3: Eigen/Katz/PageRank (10 points)

Consider the undirected graph $G = (V, E)$ shown below:



Let the $\alpha = 0.25$ and $\beta = 0.2$

The eigenvalues of the adjacency matrix A of the graph are given as:

$$\lambda = \{-1.68, -1.0, -1.0, 0.35, 3.32\}$$

- Construct the adjacent matrix A corresponding to the graph (4 points)
- Compute the Katz centrality vector C_{Katz} , defined as:

$$C_{Katz} = \beta(I - \alpha A^T)^{-1} \cdot \mathbf{1}$$

Where I is the Identity matrix and $\mathbf{1}$ is a vector of ones. (3 points)

- Why does the eigenvector centrality differ from degree centrality? Explain the idea of “being connected to important nodes makes a node important.” (3 points)

Solution:

a) Adjacency Matrix A

Node ordering: (1, 2, 3, 4, 5)

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

b)

Condition: $\alpha < \frac{1}{|\lambda_{max}|}$.

Calculation: $\frac{1}{3.32} \approx 0.301$. Since $0.25 < 0.301$, the calculation is valid.

With $\alpha = 0.25$, $\beta = 0.2$

$$C_{Katz} = 0.2 \cdot \left(\begin{bmatrix} 1 & -0.25 & -0.25 & -0.25 & 0 \\ -0.25 & 1 & -0.25 & -0.25 & -0.25 \\ -0.25 & -0.25 & 1 & -0.25 & -0.25 \\ -0.25 & -0.25 & -0.25 & 1 & 0 \\ 0 & -0.25 & -0.25 & 0 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$C_{Katz} \approx \begin{bmatrix} 1.14 \\ 1.31 \\ 1.31 \\ 1.14 \\ 0.85 \end{bmatrix}$$

c)

Why eigenvector centrality differs from degree centrality:

Degree centrality counts the number of direct connections a node has, treating all neighbors as equally important. In contrast, eigenvector centrality gives different weights to connections based on the importance of the neighboring nodes. Therefore, not all links contribute equally.

The idea that being connected to important nodes makes a node important:

Eigenvector centrality is based on a recursive principle: a node is important if it connects to other important nodes. Connections to highly influential nodes boost a node's score more than connections to less influential ones. This creates a feedback loop in which importance spreads through the network. As a result, a node with fewer but more valuable connections can be more important than a node with many low-value connections.

Question 4: Fruchterman-Reingold Layout (10 points)

Suppose four nodes A, B, C, D lie on a line at positions: $A = (0,0)$, $B = (1,0)$, $C = (3,0)$, $D = (4,0)$ and edges are (A, B) and (C, D) . Suppose $k = 2$.

In this problem, please use:

- Attractive force: $f_a(d) = \frac{d^2}{k}$
- Repulsive force: $f_r(d) = \frac{k^2}{d}$

- a) **Compute the attractive force** on edge (A, B) and the **repulsive force** between two nodes, B and C . (5 points)
- b) What will likely happen if a Fruchterman-Reingold layout is used only with attractive forces and ignores repulsive forces? (5 points)

Solution:

a)

- $A = (0,0)$, $B = (1,0)$

Distance: $d = |1 - 0| = 1$

$$f_a(1) = \frac{1^2}{2} = 0.5$$

- $B = (1,0)$, $C = (3,0)$

Distance: $d = |3 - 1| = 2$

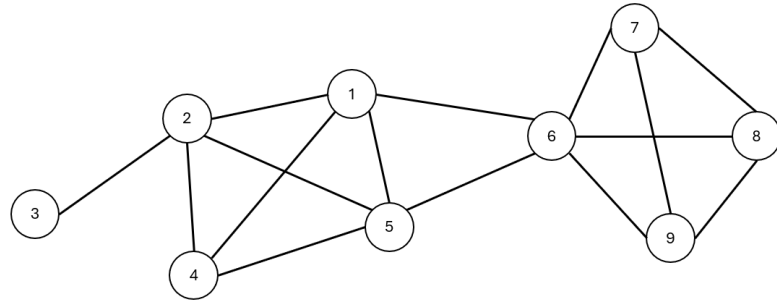
$$f_r(2) = \frac{2^2}{2} = \frac{4}{2} = 2$$

b)

The layout will **collapse** into a very compact or even single-point cluster. Without repulsive forces, there is nothing to push nodes apart, so all nodes connected by edges will keep pulling each other closer indefinitely. This results in heavy node overlap, making the visualization unreadable and losing any meaningful structural information about the network.

Question 5: Clustering Coefficient (10 points)

1. Consider an undirected graph G of nine nodes given in the following figure. (10pt)



a) Calculate the **global clustering** coefficient C_i (4 points)

$$\text{Given: } C_i = \frac{\text{\#of closed triplets}}{\text{\#of connected triplets}}$$

b) Calculate the **average clustering coefficient** $\langle C \rangle$ in the graph G (4 points)

$$\text{Given: } \langle C \rangle = \frac{1}{N} \sum_{i=1}^N C_i$$

c) Clustering coefficient is closely related to the presence of triangles in the network. Why are triangles important for understanding local structure? (2 points)

Solution:

a)

Calculate the global clustering coefficient C_i

The global clustering coefficient is the ratio of closed triplets to the total number of connected triplets (paths of length 2).

1. Calculate the total number of connected triplets (Denominator):

A connected triplet centered at a given node is calculated as $\binom{k}{2} = \frac{k(k-1)}{2}$, where k is the node's degree.

- **Node 1:** $k = 4 \Rightarrow \binom{4}{2} = 6$
- **Node 2:** $k = 4 \Rightarrow \binom{4}{2} = 6$
- **Node 3:** $k = 1 \Rightarrow \binom{1}{2} = 0$
- **Node 4:** $k = 3 \Rightarrow \binom{3}{2} = 3$
- **Node 5:** $k = 4 \Rightarrow \binom{4}{2} = 6$
- **Node 6:** $k = 5 \Rightarrow \binom{5}{2} = 10$
- **Node 7:** $k = 3 \Rightarrow \binom{3}{2} = 3$

- **Node 8:** $k = 3 \Rightarrow \binom{3}{2} = 3$
- **Node 9:** $k = 3 \Rightarrow \binom{3}{2} = 3$

Total connected triplets = $6 + 6 + 0 + 3 + 6 + 10 + 3 + 3 + 3 = 40$

2. Calculate the total number of closed triplets (Numerator):

Every unique triangle in the graph is made up of exactly 3 closed triplets. We can find the number of closed triplets by counting the total triangles and multiplying by 3.

- **Left structure triangles:** $\{1,2,4\}, \{1,2,5\}, \{1,4,5\}, \{2,4,5\}, \{1,5,6\}$ (5 triangles)
- **Right structure triangles:** $\{6,7,8\}, \{6,7,9\}, \{6,8,9\}, \{7,8,9\}$ (4 triangles)

Total triangles = 9.

Total closed triplets = $9 \times 3 = 27$

3. Final Calculation:

$$C_i = \frac{27}{40} = 0.675$$

b)

Local Coefficient Breakdown:

- **Node 1:** Neighbors $\{2, 4, 5, 6\}$. Edges between them: (2,4), (2,5), (4,5), (5,6).

$$C_1 = \frac{4}{6} = \frac{2}{3}$$

- **Node 2:** Neighbors $\{1, 3, 4, 5\}$. Edges between them: (1,4), (1,5), (4,5).

$$C_2 = \frac{3}{6} = \frac{1}{2}$$

- **Node 3:** Neighbors $\{2\}$. Since $k < 2$, no triangles can form.

$$C_3 = 0$$

- **Node 4:** Neighbors $\{1, 2, 5\}$. Edges between them: (1,2), (1,5), (2,5).

$$C_4 = \frac{3}{3} = 1$$

- **Node 5:** Neighbors $\{1, 2, 4, 6\}$. Edges between them: (1,2), (1,4), (1,6), (2,4).

$$C_5 = \frac{4}{6} = \frac{2}{3}$$

- **Node 6:** Neighbors $\{1, 5, 7, 8, 9\}$. Edges between them: (1,5), (7,8), (7,9), (8,9).

$$C_6 = \frac{4}{10} = \frac{2}{5}$$

- **Node 7:** Neighbors $\{6, 8, 9\}$. Edges between them: (6,8), (6,9), (8,9).

$$C_7 = \frac{3}{3} = 1$$

- **Node 8:** Neighbors $\{6, 7, 9\}$. Edges between them: (6,7), (6,9), (7,9).

$$C_8 = \frac{3}{3} = 1$$

- **Node 9:** Neighbors $\{6, 7, 8\}$. Edges between them: (6,7), (6,8), (7,8).

$$C_9 = \frac{3}{3} = 1$$

Sum of Local Coefficients:

$$\sum C_i = \frac{2}{3} + \frac{1}{2} + 0 + 1 + \frac{2}{3} + \frac{2}{5} + 1 + 1 + 1 = \frac{187}{30} \approx 6.233$$

Final Calculation ($N = 9$):

$$\langle C \rangle = \frac{1}{9} \times \frac{187}{30} = \frac{187}{270} \approx 0.693$$

c)

Triangles capture local cohesion: A triangle means that three nodes are mutually connected, indicating a tightly knit group in which the neighbors of a node are also connected to each other.

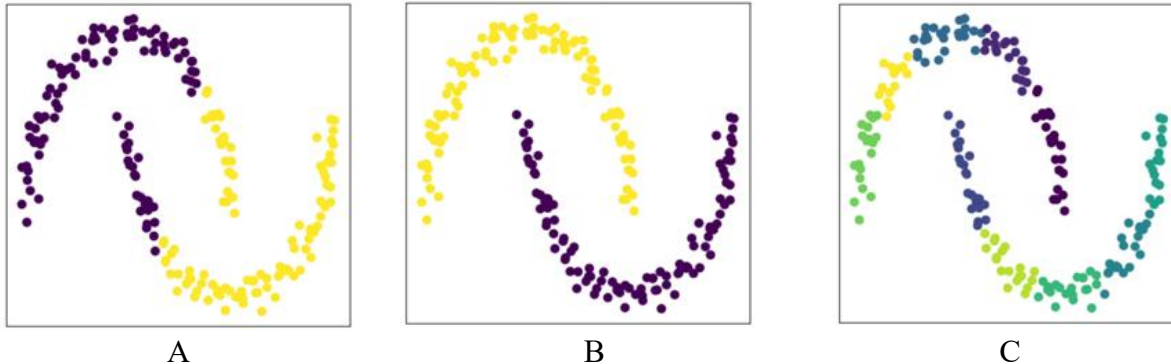
They reveal community structure: High numbers of triangles suggest the presence of clusters or communities, where nodes form dense local neighborhoods rather than random connections.

They reflect transitivity (friend-of-a-friend effect): In many real networks, if node A is connected to B and B to C, there is a higher probability that A is also connected to C. Triangles quantify this tendency.

They indicate redundancy and robustness: Multiple connections within a triangle provide alternative paths, making the network locally more resilient to node or edge removal.

Question 6: Spectral Clustering (5 points)

Given three clustering results for two moons, shown in Figures A, B, and C. Which figure most likely represents the output of spectral clustering? **Explain your choice.**



Solution:

The correct answer is **Figure B**.

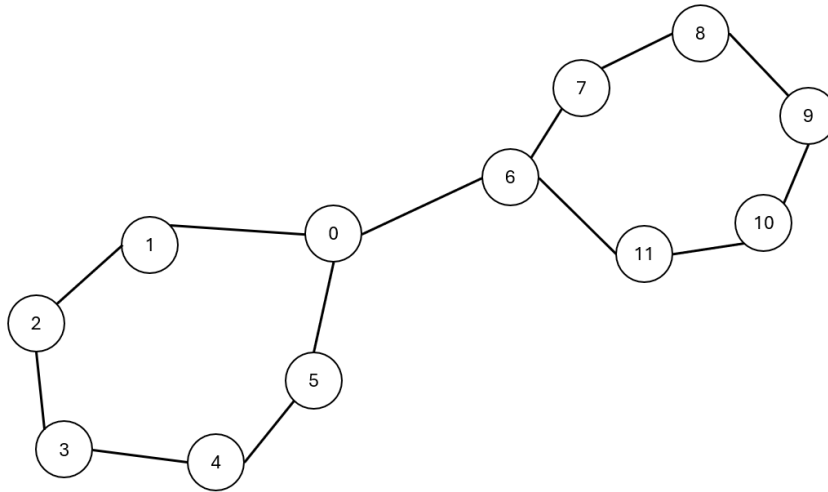
Figure A (K-Means): This result is consistent with K-Means clustering, which relies on Euclidean distance and assumes convex (spherical) clusters. It fails to capture the non-linear “two moons” structure. As a result, it incorrectly splits the data into regions that mix parts of both moons.

Figure B (Spectral Clustering): This is the expected output of spectral clustering. Spectral methods use the graph structure of the data and can capture non-linear relationships. It correctly separates the two moons into two coherent clusters that follow their curved shapes.

Figure C (Louvain, 10 communities): This result likely comes from applying the Louvain method with a setting that detects multiple communities (e.g., 10). It over-segments the data into many small clusters. While Louvain is effective for community detection, it is not constrained to find exactly two clusters, leading to fragmented results in this case.

Question 7: Min-Cut and Normalized-Cut (10 points)

Given an undirected graph G of twelve nodes, given as follows. There are two communities that exist in the graph, $A = \{0, 1, 2, 3, 4, 5\}$ and $B = \{6, 7, 8, 9, 10, 11\}$



Given:

- **Cut:** A set of edges with only one vertex in a group:

$$cut(A, B) = \sum_{i \in A, j \in B} w_{ij}$$

- **Normalized Cut**

$$NCut(A, B) = \frac{cut(A, B)}{vol(A)} + \frac{cut(A, B)}{vol(B)}$$

Where:

- $vol(A)$ is the volume of set A , defined as the sum of the degrees of all nodes within A : $vol(A) = \sum_{u \in A} d(u)$ and $d(u)$ is the degree, or sum of edge weights, connected to node u
 - $vol(B)$ is the volume of set B
- a) **Calculate min-cut** and **normalized cut** measurements of A and B (5 points)
- b) Why was the idea of **Normalized-cut** introduced? What weakness of Min-cut is N-Cut trying to solve? (5 points)

Solution:

a)

Communities:

- $A = \{0,1,2,3,4,5\}$
- $B = \{6,7,8,9,10,11\}$

From the graph, the **only edge connecting A and B** is: (0, 6)

$$\text{cut}(A, B) = 1$$

Compute volumes:

$$\text{vol}(A) = 3 + 2 + 2 + 2 + 2 + 2 = 13$$

$$\text{vol}(B) = 3 + 2 + 2 + 2 + 2 + 2 = 13$$

Compute NCut:

$$\text{NCut}(A, B) = \frac{1}{13} + \frac{1}{13} = \frac{2}{13} \approx 0.154$$

b)

Weakness of Min-cut:

Min-cut tends to produce unbalanced partitions, often isolating a small set of nodes (or even a single node), because it minimizes the number of cut edges.

- It focuses only on minimizing the number (or weight) of edges between groups.
- It ignores the size or volume of the resulting partitions.

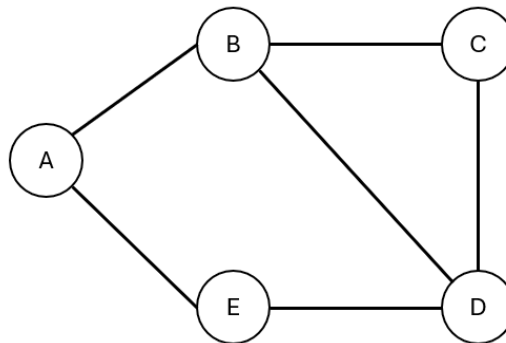
What N-cut solves:

- Normalized-cut adds a normalization term based on the size (volume) of each partition.
- It penalizes cuts that separate very small subsets from the rest of the graph.
- It encourages balanced and meaningful partitions.

Key idea:

Instead of just minimizing connections between groups, N-cut minimizes the **fraction of edges leaving each group relative to the total connections within that group**, leading to more interpretable community structures.

Question 8: Edge Betweenness (10 points)



Given:

$$c_B(e) = \sum_{s \neq t} \frac{\sigma_{st}(e)}{\sigma_{st}}$$

Where:

- s and t are pairs of nodes in the graph.
 - σ_{st} is the total number of shortest paths from node s to node t .
 - $\sigma_{st}(e)$ is the number of those shortest paths that explicitly pass through edge e .
- a) Compute the **edge betweenness** of edge (A, B) (7 points)
- b) In a transportation network, one road connects two large cities, while many roads lie inside each region. Which types of roads are likely to have higher edge betweenness? Explain your answer in detail. (3 points)

Solution:

a)

Pair {A, B}: * Shortest path: A-B

Total shortest paths: 1

Paths through (A, B): 1

Contribution: $1 / 1 = 1$

Pair {A, C}:

Shortest path: A-B-C

Total shortest paths: 1

Paths through (A, B): 1

Contribution: $1 / 1 = 1$

Pair {A, D}:

Shortest paths: A-B-D and A-E-D

Total shortest paths: 2

Paths through (A, B): 1 (specifically, A-B-D)

Contribution: $1 / 2 = 0.5$

Pair {E, B}:

Shortest paths: E-A-B and E-D-B

Total shortest paths: 2

Paths through (A, B): 1 (specifically, E-A-B)

Contribution: $1 / 2 = 0.5$

For all other possible pairs in the graph ($\{A, E\}$, $\{B, C\}$, $\{B, D\}$, $\{C, D\}$, $\{C, E\}$, and $\{D, E\}$), the shortest paths do not traverse the edge (A, B), meaning their contribution to the sum is **0**.

$$C_B(A, B) = 1 + 1 + 0.5 + 0.5 = 3$$

b)

- **Road connecting the two large cities (bridge between regions):**

This road will have high edge betweenness. Many shortest paths between nodes in different cities must pass through this road. It acts as a bottleneck or bridge, concentrating inter-city traffic. Therefore, it lies on many shortest paths across the network.

- **Roads inside each region (intra-city roads):**

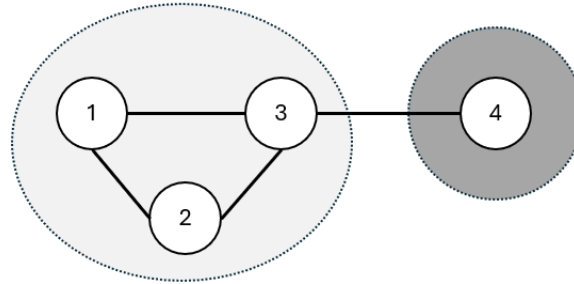
These roads will generally have lower edge betweenness. There are many alternative routes within a city, so shortest paths span multiple edges. No single road dominates the flow of shortest paths.

- **Conclusion:**

Edge betweenness is highest for critical connectors between communities (cities), rather than for roads within densely connected regions.

Question 9: Modularity (10 points)

Consider an undirected graph G of four nodes as illustrated in the following figure. Knowing that, there exist two communities $A = \{1, 2, 3\}$ and $B = \{4\}$



Given:

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{d_i d_j}{2m} \right) \cdot \delta(v_i, v_j)$$

Where:

- $\delta(v_i, v_j) = \begin{cases} 1 & \text{if } v_i \text{ and } v_j \text{ are in the same community.} \\ 0 & \text{otherwise.} \end{cases}$
 - m is the number of total edges
 - A is the adjacency matrix of G
 - d_i is the degree of the node v_i
- a) Calculate the **Modularity** of the two communities (6 points)
 - b) In a social network, modularity is used to evaluate a proposed division of users into communities. What would a high modularity value suggest about the social structure? (4 points)

Solution:

a)

$$Q = 1/8 *$$

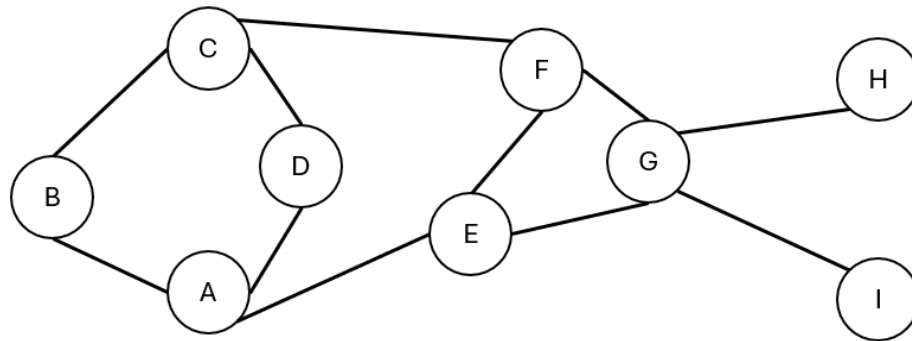
	v=1	v=2	v=3	v=4
u=1	$(0 - (2*2) / 8) * 1$	$(1 - (2*2) / 8) * 1$	$(1 - (2*3) / 8) * 1$	$(0 - (2*1) / 8) * 0$
u=2	$(1 - (2*2) / 8) * 1$	$(0 - (2*2) / 8) * 1$	$(1 - (2*3) / 8) * 1$	$(0 - (2*1) / 8) * 0$
u=3	$(1 - (3*2) / 8) * 1$	$(1 - (3*2) / 8) * 1$	$(0 - (3*3) / 8) * 1$	$(1 - (3*1) / 8) * 0$
u=4	$(0 - (1*2) / 8) * 0$	$(0 - (1*2) / 8) * 0$	$(1 - (1*3) / 8) * 0$	$(0 - (1*1) / 8) * 1$

$$= -0.03125$$

b)

A **high modularity value** suggests that the network has a **clear and strong community structure**. Users within the same community are closely connected, forming tight-knit groups. Connections between different communities are relatively sparse. In other words, the proposed division captures real social groupings, such as friend circles or shared-interest groups, rather than random splits in the network.

Use the following graph to answer questions 10 and 11



Question 10: Jaccard Coefficient (5 points)

$$\text{Given: } J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

- Calculate the **Jaccard coefficient** of the non-edge pair (A, C) (4 points)
- Why might the Jaccard Coefficient be more informative than simply counting the number of common neighbors between two nodes? (1 points)

Solution:

a)

Neighbors:

$$N(A) = \{B, D, E\}$$

$$N(C) = \{B, D, F\}$$

$$\text{Common neighbors: } N(A) \cap N(C) = \{B, D\}$$

$$\text{Jacc_coef}(A, C) = \frac{|\{B, D\}|}{|\{B, D, E, F\}|} = \frac{2}{4} = \frac{1}{2}$$

- The Jaccard Coefficient is often more informative because it *normalizes the number of common neighbors by the total number of neighbors for both nodes*. Counting common neighbors alone can be misleading. Nodes with many connections are more likely to share neighbors by chance, even if they are not really similar. The Jaccard Coefficient addresses this by considering the proportion of shared neighbors rather than the total count. This approach provides a fairer and more meaningful measure of similarity, especially when comparing nodes with very different degrees.

Question 11: Adamic-Adar Index (5 points)

$$\text{Given: } A(u, v) = \sum_{w \in N(u) \cap N(v)} \frac{1}{\log |N(w)|}$$

Where:

- $N(u) \cap N(v)$: The common neighbors between the nodes u and v
 - w : A specific common neighbor in that set
 - $|N(w)|$: The total number of neighbors of the common neighbor w
- a) Calculate the **Adamic-Adar Index** of the non-edge pair (C, E) (4 points)
- b) What does a high Adamic-Adar Index between two nodes suggest about their relationship in the network? (1 points)

Solution:

a)

Neighbors:

$$N(E) = \{A, F, G\}$$

$$N(C) = \{B, D, F\}$$

$$\text{Common neighbors: } N(E) \cap N(C) = \{F\}$$

$$A(A, C) = \frac{1}{\log(3)} = 0.91$$

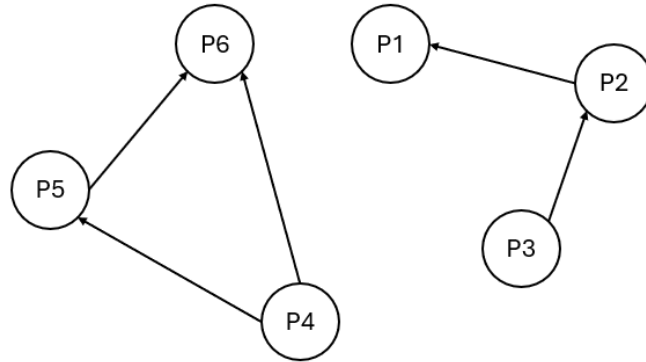
b)

A high Adamic-Adar Index between two nodes means they share mutual connections that are relatively uncommon or not widely connected to others, making those shared neighbors more meaningful. Unlike simply counting common connections, this measure gives greater weight to rare, low-degree neighbors, since they are less likely to be coincidental. As a result, a higher score suggests a stronger similarity between the two nodes and a higher likelihood that they are closely related or will form a connection, such as in a social network where people with unique mutual friends are more likely to become connected.

Intuition: If you and another person both know a rare, specialized person, it is a stronger sign you are connected than if you both know a celebrity who is connected to millions.

Question 12: Eigen/Katz/PageRank (5 points)

Consider this mini-internet:



Given: **Original PageRank Formula** as:

$$PR(A) = (1 - d) + d \sum_{i=1}^n \frac{PR(T_i)}{C(T_i)}$$

Where:

- $PR(A)$: The PageRank of the target page A .
 - $T_1, T_2, \dots, T_n$: The pages that have an outbound link pointing to page A .
 - $PR(T_i)$: The PageRank score of the linking page T_i .
 - $C(T_i)$: The total number of outbound links on page T_i .
 - d : The damping factor. (In this assignment, use $d = 0.85$)
- a) **Calculate the PageRank** of all pages for **2 iterations**. Consider the initial value for each node is 1. (4 points)
- b) If Katz centrality already improves the eigenvector centrality, why was PageRank still needed? What additional problems does PageRank address? (1 points)

Solution:

a)

$d = 0.85$ is the damping factor, making the base rank $1 - 0.85 = 0.15$.

Initial:

$$PR^{(0)}(v) = 1$$

Out-degrees:

$$C(P4) = 2, C(P5) = 1, C(P3) = 1, C(P2) = 1$$

Iteration 1

- P1: $PR = 0.15 + 0.85(1) = 1.0$
- P2: $PR = 0.15 + 0.85(1) = 1.0$
- P3: $PR = 0.15$
- P4: $PR = 0.15$
- P5: $PR = 0.15 + 0.85 \cdot \frac{1}{2} = 0.575$
- P6: $PR = 0.15 + 0.85 \left(\frac{1}{2} + 1 \right) = 0.15 + 0.85(1.5) = 1.425$

Iteration 2

- P1: $PR = 0.15 + 0.85(1.0) = 1.0$
- P2: $PR = 0.15 + 0.85(0.15) = 0.2775$
- P3: $PR = 0.15$
- P4: $PR = 0.15$
- P5: $PR = 0.15 + 0.85 \cdot \frac{0.15}{2} = 0.21375$
- P6: $PR = 0.15 + 0.85 \left(\frac{0.15}{2} + 0.575 \right) = 0.7025$

Node	Iteration 0	Iteration 1	Iteration 2
P3	1	0.15	0.15
P2	1	1.0	0.2775
P1	1	1.0	1.0
P4	1	0.15	0.15
P5	1	0.575	0.21375
P6	1	1.425	0.7025

b)

Although Katz centrality is an improvement over eigenvector centrality because it considers both direct and indirect connections with some reduction, it has limitations in certain types of networks, especially directed ones like the web.

Problem of sinks (dangling nodes):

Katz centrality does not handle nodes with no outgoing edges well. In contrast, PageRank uses a damping factor that redistributes importance, preventing the rank from becoming stuck in these nodes.

Handling directed networks more effectively:

PageRank is designed for directed graphs, where link direction, like hyperlinks, is important. It models a random surfer who follows links, making it better suited for web-like structures.

Normalization of influence:

PageRank distributes a node's importance among its outgoing links. This prevents nodes with many outgoing edges from unfairly passing on too much influence.

Global stability and interpretability:

The damping or random-jump mechanism helps ensure convergence and provides a more reliable measure of importance in large, complex networks.